

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

**General Certificate of Education Ordinary Level
MATHEMATICS 4004/2 PAPER 2 - November 2022**

O-Level | Time: 2 hours 30 minutes | 100 Marks | Section A (52) + Section B (48) | Calculator allowed

Centre Number _____ Candidate Number _____ Name _____

SECTION A (52 Marks) - Answer ALL questions in this section

1. (a) Simplify $4x - 2(x+3)$ [2] (b) Solve $(x+2)/3 = 4$ [2] (Algebraic manipulation)

2. (a) Share \$120 in ratio 2:3 [2] (b) Convert US\$50 to ZWL at 1:322 [2] (Number Theory & Currency)

3. (a) Describe region in Venn diagram [2] (b) Find HCF and LCM [2] (Set Theory)

4. (a) Calculate circumference circle radius 7 [2] (b) Find exterior angle of regular octagon [2] (Geometry)

5. Construct quadrilateral ABCD with AB=6cm BC=5cm angle 90 degrees using ruler and compasses [6] (Geometrical Construction)

6. Hire Purchase: Deposit \$20 + $12 \times \$5$. Find cash price vs HP [4] (Commercial Maths)

SECTION B (48 Marks) - Answer any FOUR questions from this section

7. Algebraic Fractions & Quadratics: Simplify $(x^2-4)/(x^2+3x+2)$ and form quadratic from area 24 [12]

8. Similar Figures: Area ratio triangles 9:16, find sides, cylinder surface area [12]

9. Kinematics: Speed-time graph, acceleration, distance [12]

10. Variation: y varies directly as x^2 , find variation equation [12]

11. Trigonometry & Vectors: Non-right triangle, column vectors, midpoint [12]

12. Statistics: Frequency table, ogive, median [12]
